Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat
Department of Artificial Intelligence
Intelligent Multi-Agent and Expert Systems (AI401), Semester VII

Instructions

1. Save your file as LAB_No_Roll_No.doc (e.g. LAB_7_U21AI001.doc).
2. Paste a snapshot of each step followed along with the result(s).
3. Record your observations and comments after the results.

Experiment 7

Negotiation Strategies — Contract Net and Alternating Offers in FIPA-ACL

Aim

To simulate agent negotiation using FIPA-ACL messages under two protocols — the Contract Net, in which a manager awards each task to the best proposal received, and alternating offers, in which two agents concede towards each other under a deadline — and to measure how the outcome changes with the strategy each agent adopts.

Theory

An agent communication language fixes the envelope rather than the content: a FIPA-ACL message carries a performative stating what the sender is doing with the message, together with sender, receiver, content, a conversation identifier and the protocol being followed. The Contract Net uses five of them in a fixed order. The manager issues a cfp to every candidate; each candidate replies with propose, naming a price, or with refuse; the manager returns accept-proposal to one and reject-proposal to the rest; the winner closes the conversation with inform-done. The conversation identifier is what lets several of these run at once without their messages being confused, and it is the only thing distinguishing two otherwise identical proposals.

The protocol fixes the moves; the strategy is what an agent chooses within them. In the Contract Net the choice is the price to propose, and since a bid equal to cost earns nothing, a self-interested agent marks it up until the risk of losing outweighs the margin — this is a first-price auction, and it does not reward honesty. In bilateral bargaining the choice is a concession schedule: both sides open at their most favourable price and move towards their reservation value as a deadline approaches, a boulware agent conceding almost nothing until the end and a conceder giving ground early. Neither strategy is good in isolation, because the agreement price depends on both schedules at once, which is what the exercises measure.

Algorithm

Algorithm 1 — Contract Net under FIPA-ACL

1. Represent an ACL message as a record of performative, sender, receiver, content, conversation identifier and protocol. Place the agents and tasks on a grid as in Experiment 6, take an agent’s cost for a task to be the Manhattan distance, and give every agent a reservation above which it refuses rather than proposes.
2. For each task in turn, open a conversation: send a cfp to every agent that is still free, collect a propose carrying the bid or a refuse from each, send accept-proposal to the lowest bidder and reject-proposal to the others, and record the winner’s inform-done. Print the full message transcript, the awards, the total the manager pays and the number of messages exchanged.

Algorithm 2 — Alternating offers with time-dependent concession

1. Give the buyer and the seller an opening price, a reservation price, a shared deadline in rounds and a concession parameter; define the offer of each side at round t as its opening price moved towards its reservation price by a fraction that depends on t and on that parameter, so that a value below one concedes late and a value above one concedes early.
2. In each round exchange propose messages carrying the current offers, and have a side send accept-proposal as soon as the offer it has received is at least as good as the offer it was about to make; if the deadline passes with no agreement, close the conversation with cancel. Report the transcript, the agreed price, the round of agreement and the utility each side obtains.

Procedure and Result

Paste the snapshots of the steps followed and the output produced by the program below, including the cost matrix, the Contract Net transcript for at least one complete conversation with every performative visible, the awards with the total paid and the message count, and the offer sequence and agreement of one bargaining run.

Observations

Comment on what a refuse costs the manager compared with a losing propose, on whether announcing the tasks in a different order changes what the manager pays and why a commitment once accepted cannot be undone, on which side gains when the two concession rates differ, and on what the deadline contributes to the bargaining outcome.

Exercises

1. Run the bargaining protocol for every pairing of a boulware, a linear and a conceder strategy on each side, and report the agreed price, the round of agreement and both utilities for all nine pairings. State which pairing settles earliest, which gives the most lopsided split, and whether two boulware agents reach agreement before the deadline at all.
2. Let one contractor inflate every proposal by a markup factor while the others bid truthfully. Sweep the factor and report the tasks that contractor wins, the number of tasks served, the total the manager pays and the contractor’s profit. Identify the markup at which it stops winning, and state what this implies about bidding honestly when the lowest bid is paid.